

CASM: A CONTEXT-AWARE SEMI-MARKOV ALTERNATIVE TO DBN POST-PROCESSING FOR BEAT TRACKING

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ABSTRACT

The last stage of beat tracking must turn dense neural activations into a discrete and musically coherent event sequence. Direct peak picking preserves local evidence but can fragment the pulse, whereas dynamic Bayesian networks (DBNs) restore continuity through globally configured tempo and meter assumptions that need not transfer to difficult music. We introduce CASM, a plug-in, context-aware semi-Markov decoder over a sparse graph of activation-supported beat candidates. CASM estimates local periodicity and its ambiguity from the current activation sequence; this ambiguity jointly controls the strength and width of each event-to-event duration potential. It therefore commits strongly when the pulse is clear and relaxes toward the neural evidence when competing metrical levels are plausible. A beat-count safeguard and a beat-synchronous meter decoder make this behaviour deterministic and auditable. One globally calibrated CASM configuration is applied without backbone retraining or track-specific BPM and meter settings to Beat This, MSCNN-lite, and TCN activations on GTZAN and SMC. The results position CASM as a conservative F1-continuity operating point rather than a universal replacement for DBN. An exploratory calibration-scale study further shows metric-dependent saturation and decreasing sensitivity to fold composition.

Index Terms— beat tracking, downbeat tracking, semi-Markov decoding, post-processing, local periodicity

1. INTRODUCTION

The final decoder in a beat tracker is deceptively consequential. A neural network can assign high framewise probability to genuine beats while still producing duplicates, gaps, metrical-level switches, or downbeats that are inconsistent with its beat grid. Event-level F-measure rewards accurate local placement, whereas continuity measures additionally ask whether one metrical interpretation persists through time. This creates a practical tension: weak decoding respects a capable network but may yield an unstable clock; strong decoding restores a clock but can overwrite correct, expressive, or unusual evidence.

One prevalent structured decoder is the dynamic Bayesian network (DBN) [1]. Its latent tempo, position, and meter states are valuable precisely because they can repair noisy activations. The permitted tempo support, meter set, and transition stiffness, however, are configured globally. The posterior adapts its state path to the input, but the form and

strength of the temporal assumptions do not become more cautious when the input itself reveals half/double-tempo ambiguity. Re-optimizing those settings on a target corpus is legitimate development-set model selection, yet a test- or target-informed recipe is no longer evidence of out-of-the-box transfer. Recent post-processing-free trackers show that strong networks can avoid some DBN errors [2, 3], but minimal peak picking can sacrifice sequence continuity. Recent failure analysis on the deliberately difficult SMC corpus makes this trade-off especially visible [4].

We ask whether the temporal prior itself can listen to the current activation. Our answer is CASM, a Context-Aware Semi-Markov post-processor. CASM constructs a sparse graph whose nodes are activation maxima and whose edges are complete inter-beat segments. At both endpoints of an edge, it estimates a local period and a best-versus-competing-period margin. Clear periodic evidence creates a narrow, strong duration potential; ambiguous evidence simultaneously weakens and broadens that potential. CASM is consequently an automatic correction layer in a precise, limited sense: after one global calibration, it requires neither track metadata nor per-track decoder adjustment, while its constraints remain input-dependent at inference.

Our contributions are threefold:

- We formulate maximum-score beat decoding on a sparse semi-Markov graph of activation-supported candidates. Each duration potential uses both endpoint contexts and is conditioned on their ambiguity.
- We connect beat decoding to a beat-synchronous variable-meter decoder and two explicit safeguards, yielding a deterministic plug-in that does not retrain the upstream tracker.
- We study one frozen operating point across three different activation backbones and analyse how the amount and composition of decoder-calibration data affect transfer, while separating exploratory selection from clean evaluation claims.

2. RELATED WORK

2.1. From activations to event sequences

Dynamic programming has long combined onset evidence with inter-beat regularity [5]. Korzeniowski et al. proposed a Bayesian beat-position extractor around a dominant beat interval [6]; its implementation is exposed by madmom under the historical name CRFBeatDetectionProcessor, al-

though the paper is not the learned linear-chain CRF of Fillon et al. [7]. The learned CRF branch was extended to downbeats [8] and, in 2019, to skip-chain musical structure and Afro-Latin microtiming [9, 10]. Resonating comb filters form a different branch: they estimate a dominant global tempo from recurrent activations [11], after which events are aligned to that periodicity. The 2019 multi-task tempo-beat model [12] extended neural training rather than this comb-filter event decoder, and retained structured inference. DBNs instead enumerate tempo-phase-meter states and decode their most likely trajectory [1, 13]; they remain common in recent high-performing systems.

2.2. Local, segmental, and modern alternatives

Reliability-informed tracking established the principle that temporal commitment should depend on evidence quality [14]. More recently, a compact one-dimensional semi-Markov state space represented time-varying rhythmic jumps for causal joint tracking [15]. Particle filters have long maintained online tempo hypotheses [16], and BeatNet/BeatNet+ extend that branch to joint real-time rhythm analysis [17, 18]. PLPDP is CASM’s closest conceptual predecessor: it derives predominant local pulse information and uses dense-frame dynamic programming for expressive tempo [19]. Its real-time continuation improves controllability [20], and dPLP makes local-pulse estimation differentiable [21].

CASM does not claim novelty for local periodicity, dynamic programming, or semi-Markov inference separately. Unlike PLPDP’s dense frame grid, it scores only segments joining retained activation maxima and conditions each segment on the local period and ambiguity at both endpoints. Ambiguity modulates both penalty strength and tolerance, rather than merely supplying a preferred period. Unlike the one-dimensional causal semi-Markov model, CASM is an off-line sparse event graph with explicit fallback behaviour. Unlike a DBN, it does not maintain a dense tempo-phase state lattice; unlike a semi-CRF [22], its potentials are not learned or normalized.

Another response to difficult rhythm is to adapt the tracker with user annotations [23–27]. That line is complementary: it deliberately imports target-specific supervision. CASM instead recomputes its constraint from unlabelled test activations and leaves upstream weights fixed. At the other extreme, recent work reformulates the prediction task itself [28]; masked diffusion learns coherent sequences through iterative refinement [29], while BeatFCOS predicts intervals as objects with Soft-NMS [30]. Knowledge-driven metrical hypotheses have also been introduced during self-supervised pre-training [31]. These are modern alternatives to traditional post-processing, but require a changed or retrained predictor; CASM targets the still-useful deployment setting in which only a frozen activation interface is available.

3. CASM: CONTEXT-AWARE SEMI-MARKOV DECODING

3.1. Candidate graph and segmental objective

Let a frozen tracker output beat and downbeat logits z_t^b and z_t^d at F frames per second, and let $a_t = \text{sigmoid}(z_t^b)$. We retain one-dimensional local maxima, including valid boundary

maxima, whose probability exceeds θ_c . The ordered candidate set is $\mathcal{C} = \{t_1, \dots, t_N\}$. A directed edge (i, j) spans the complete candidate-to-candidate interval

$$\Delta_{ij} = (t_j - t_i)/F$$

and is admissible when $\Delta_{ij} \in [60/B_{\max}, 60/B_{\min}]$. CASM never inserts an event away from a retained neural maximum; it selects a sparse path $\pi = (i_1, \dots, i_K)$ by maximizing

$$\mathcal{S}(\pi) = \sum_{k=1}^K e_{i_k} - \sum_{k=2}^K D_{i_{k-1}, i_k}. \quad (1)$$

Here e_i is clipped logit evidence and D_{ij} scores the duration of a whole segment. This variable-duration edge potential is the precise semi-Markov component; Eq. (1) is a maximum-score decoder, not a generative hidden semi-Markov model.

3.2. Local period and ambiguity

For each candidate i , we evaluate integer lags $\ell \in \mathcal{L} = \{\ell_{\min}, \dots, \ell_{\max}\}$ in a centred local window $\mathcal{W}_i$. Zero-padded normalized autocorrelation gives

$$q_i(\ell) = \frac{\langle a_u a_{u-\ell} \rangle_{u \in \mathcal{W}_i}}{\sqrt{\langle a_u^2 \rangle_{u \in \mathcal{W}_i} \langle a_{u-\ell}^2 \rangle_{u \in \mathcal{W}_i} + \epsilon_q}} \left(\frac{\ell_{\min}}{\ell} \right)^\beta, \quad (2)$$

where the last factor is a mild fast-tempo bias. We take $\ell_i^* = \arg \max_{\ell} q_i(\ell)$ and $\tau_i = \ell_i^*/F$. The runner-up excludes the best lag and its two neighbours, exposing competing nearby and metrical-level periodicities. Their separation is

$$c_i = \text{clip}_{[0,1]} \left[\frac{q_i(\ell_i^*) - \max_{\ell: |\ell - \ell_i^*| > 2} q_i(\ell)}{|q_i(\ell_i^*)| + \epsilon_c} \right]. \quad (3)$$

We call c_i a period margin, not calibrated correctness. If at least five candidates exist, both τ_i and c_i are median-filtered over five successive candidates.

3.3. Ambiguity-conditioned duration inference

For edge (i, j) , CASM combines the endpoint contexts geometrically: $\bar{\tau}_{ij} = \sqrt{\tau_i \tau_j}$ and $c_{ij} = \sqrt{c_i c_j}$. Its duration cost is

$$D_{ij} = \frac{\lambda c_{ij}}{2} \left[\frac{\log(\Delta_{ij}/\bar{\tau}_{ij})}{\sigma_0 + (1 - c_{ij})\sigma_u} \right]^2. \quad (4)$$

The log ratio treats proportional timing errors symmetrically. More importantly, c_{ij} plays two coupled roles. A separated local-period peak increases the multiplier and narrows the scale, enforcing a clear pulse. Competing periods reduce the multiplier and widen the scale, allowing the path to follow activation evidence instead of confidently imposing the wrong metrical level.

With $e_j = \text{clip}(z_{t_j}^b/T, -L, L)$, the Viterbi-style recurrence is

$$V_j = e_j + \max \left\{ 0, \max_{\substack{i < j \\ (i,j) \text{ admissible}}} (V_i - D_{ij}) \right\}. \quad (5)$$

The zero branch permits a new path after an unfavourable prefix, and backtracking starts from $\arg \max_j V_j$. In parallel, a Direct path is formed by seven-frame max pooling, a strict zero-logit threshold, and one-frame peak deduplication,

matching Beat This’s minimal decoder [3]. If the number of structured events divided by the number of Direct events lies outside $[r_{\min}, r_{\max}]$, the Direct path is returned. Empty candidate input returns an empty sequence. The count test is a deterministic safety gate, not a learned router.

3.4. Beat-synchronous downbeats

Let the decoded beat grid be $\hat{\mathcal{B}} = \{\hat{b}_n\}$. CASM scores possible downbeats only at these beat locations. For a bar length m in the allowed meter set $\mathcal{M}$, define $g_n = \text{clip}(z_{\hat{b}_n}^d/T_d, -L, L) + \rho$. A variable-meter recurrence connects successive downbeats:

$$H_{n,m} = g_n + \max_{m' \in \mathcal{M}} [H_{n-m,m'} - \eta \mathbf{1}(m' \neq m)]. \quad (6)$$

Initial states are permitted near the beginning and termination near the end, so a decoded path can change bar length while paying switch cost η . Separately, Direct downbeat maxima are snapped to $\hat{\mathcal{B}}$. The meter path is retained only if its one-to-one event F-measure against this snapped path is at least γ under a 70-ms tolerance; otherwise CASM returns the snapped path. Consequently every reported downbeat remains on the selected beat grid.

4. EXPERIMENTS

4.1. Data, backbones, and metrics

We evaluate on GTZAN ($N = 993$) using the annotations and exclusions of Beat This [3], and on the deliberately difficult SMC corpus ($N = 217$) [32]; SMC provides beat but not downbeat annotations. We use three 50-Hz activation backbones: the Beat This spectrogram Transformer [3], the lightweight multi-scale convolutional backbone MSCNN-lite [33], and the TCN of [34]. The decoder receives only their framewise beat and downbeat logits.

For GTZAN, final0, final1, and final2 are three independently trained full-data backbone seeds. Their training excludes GTZAN but includes SMC. We first macro-average over the same 993 tracks and then average the three seed-level scores. For SMC, using those full-data checkpoints would leak training examples; we instead use the corresponding held-out checkpoint for each of eight folds. Seven folds contain 27 tracks and the eighth contains 28; we average their fold-level macro scores. CASM itself is deterministic and has neither a neural checkpoint nor a random seed.

Following the established protocol, the first 5s are trimmed and mir_eval computes F1 with ± 70 ms tolerance, CMLt, and AMLt [35]. CMLt requires a continuously correct metrical level; AMLt additionally accepts the standard off-beat and half/double-level interpretations. Seed variation on GTZAN and fold variation on SMC have different replication units and are descriptive, not confidence intervals.

4.2. Decoder baselines

Direct is the minimal peak picker defined in Sec. 3. DBN is the conventional joint madmom configuration: meters $\{3, 4\}$, 55–215 BPM, transition weight 100, observation weight 16, and threshold 0.05 [1]. DBN-SMC-opt is deliberately diagnostic. Its sequential beat and bar stages were selected from 753 configurations on the TCN SMC 8-fold OOF activations,

maximizing beat F1 with CMLt/AMLt tie-breaks, and only then frozen for the three backbones and GTZAN. It uses 50–215 BPM, 30 tempo states, transition/observation weights 7/20, threshold 0.3, meters $\{3, 4\}$, bar-observation weight 100, and meter-switch probability 10^{-7} . It is neither a clean SMC test nor a published SMC-specific recipe from [34].

The CRF row uses madmom’s historically named CRF-BeatDetectionProcessor [6] at 30–300 BPM with interval width 0.18; it should not be confused with the learned CRF in [7]. CombFilter uses madmom’s global tempo/iterative-alignment processor [11]. CRF and CombFilter decode downbeat activations at 8–100 BPM and snap their outputs to their respective beat grids.

4.3. Frozen operating point and scale analysis

Every CASM row in Sec. 5 uses exactly one explicitly loaded Frozen-4F configuration:

$$\begin{aligned} F &= 50, \quad \theta_c = 0.03, \quad (B_{\min}, B_{\max}) = (30, 300), \\ |\mathcal{W}| &= 8 \text{ s}, \quad \beta = 0.15, \quad (T, L) = (2, 6), \\ (\lambda, \sigma_0, \sigma_u) &= (4, 0.12, 0.40), \quad [r_{\min}, r_{\max}] = [0.85, 1.8], \\ \mathcal{M} &= \{2, \dots, 7\}, \quad (T_d, \eta, \rho, \gamma) = (4, 3, 0, 0.6). \end{aligned}$$

Its scalar settings were selected on the union of SMC folds $\{2, 7, 1, 3\}$ and then reused for all tracks and all backbones. Thus context-aware means input-conditioned inference, not test-time parameter updates: CASM has validation-selected parameters, but no annotated beat, inter-beat-interval, BPM, or meter input at inference.

The choice to foreground 4F was made after inspecting the calibration sensitivity analysis and is therefore exploratory. The seven-fold configuration is the only comparator locked before GTZAN inspection; it differs from 4F only in $\sigma_0 = 0.15$. For the scale study, SMC fold 0 ($N = 27$) is never used for selection. From folds 1–7 we enumerate every 1-, 2-, and 4-fold calibration subset (7, 21, and 35 combinations) plus the seven-fold union, and evaluate all locked configurations on the same SMC fold 0 and fixed GTZAN/final1 panel. The latter panel is test-conditioned and is used only to describe sensitivity, not to claim a held-out optimum.

5. RESULTS AND DISCUSSION

5.1. A single operating point across backbones

Table 1 is best read as a transfer and trade-off table, not a winner count. At its displayed precision, the same CASM setting produces positive changes from Direct across the three backbones, two corpora, and all available metrics. Its characteristic behaviour is conservative: it preserves local event placement while recovering part of the continuity absent from peak picking. DBN produces larger continuity changes in several GTZAN downbeat conditions, but those changes are frequently accompanied by lower beat or downbeat F1 and are less stable on SMC. The SMC-selected DBN also changes its trade-off when transferred back to GTZAN. We therefore do not claim that CASM dominates DBN, CRF, or comb filtering. The evidence supports a reusable operating point whose corrections depend on the input rather than a new target-specific recipe.

Table 1. Post-processor comparison. Direct rows are absolute percentages; all following rows are percentage-point changes from the Direct row of the same backbone. CASM uses one Frozen-4F configuration. DBN-SMC-opt is a target-selected diagnostic, not a clean SMC result.

Method	GTZAN (Beat / Downbeat)						SMC (Beat)		
	F1	CMLt	AMLt	F1	CMLt	AMLt	F1	CMLt	AMLt
BeatThis (direct) [3]	89.1	79.8	89.8	78.3	67.2	79.1	62.7	51.4	61.0
w. CASM	+0.2	+0.4	+0.6	+0.6	+3.2	+6.0	+0.3	+2.4	+3.3
w. DBN	-1.0	+0.7	+1.3	-0.9	+6.2	+8.7	-5.2	-3.9	+1.5
w. DBN, SMC-opt	-0.5	-0.9	+0.2	-1.5	-0.6	+1.1	-1.6	+0.4	+3.5
w. CombFilter [11]	-4.0	-3.6	+0.1	-3.3	+3.2	+7.6	-7.1	+1.1	+3.1
w. CRF [6]	-0.6	-0.3	+1.4	-1.4	+1.9	+5.5	-1.2	+1.4	+1.2
MSCNN-lite (direct) [33]	86.1	73.4	79.9	70.8	53.1	63.1	57.2	42.0	48.4
w. CASM	+0.4	+1.5	+2.4	+1.7	+6.8	+11.7	+0.5	+3.3	+4.4
w. DBN	-0.6	+4.0	+7.2	+1.8	+15.8	+21.2	-4.9	+0.3	+3.9
w. DBN, SMC-opt	-1.1	+0.2	+5.9	-0.9	+9.5	+14.6	+0.3	+3.5	+6.8
TCN (direct) [34]	87.1	73.8	81.7	62.4	24.6	59.9	52.9	30.1	36.0
w. CASM	+0.3	+1.3	+1.6	+2.3	+19.4	+14.0	+0.3	+3.8	+5.2
w. DBN	-2.6	-0.7	+5.0	+4.9	+23.2	+22.9	-0.9	-7.0	+2.3
w. DBN, SMC-opt	-1.9	-0.8	+4.4	-0.2	+8.9	+9.3	+1.7	+4.5	+8.2

5.2. Context among recent trackers

Table 2 answers a narrower question than Table 1: whether an accessible modern backbone remains competitive when its minimal decoder is replaced. We could reproduce a compatible activation interface only for Beat This, so the table does not imply that CASM improves HingeNet, BeatFM, BeatMamba, BeatKAN, or MDM. Their rows retain their authors’ training and decoding pipelines. This distinction also clarifies CASM’s practical niche. MDM and BeatFCOS alter the learned predictor to produce coherent sequences or intervals [29,30]; CASM can instead be attached to already trained logit-producing models. Testing the same decoder on future systems requires authors to expose framewise activations, not only final event times.

5.3. What calibration-scale sensitivity shows

The fixed-panel sensitivity analysis does not show that 4F is optimal, nor that more data hurts machine learning. CASM learns no network weights here: the independent variable is the amount of data used to select global decoder scalars. On the held-out SMC panel, the family centres of F1, CMLt, and AMLt rise from 1F to 7F. On GTZAN, beat F1 and downbeat F1/CMLt peak at 2F, while beat CMLt/AMLt and downbeat AMLt continue to 7F. Hence the defensible conclusion is metric-dependent diminishing returns rather than a universal best scale.

There is nevertheless a useful stability result. From 1F to 2F to 4F, the across-combination population spread decreases in all nine panels: more calibration data makes the selected decoder less sensitive to which SMC folds are present, even when the external point estimate does not improve mono-

tonically. We retain 4F as a balanced, post-hoc operating point for the present analysis. Its near-identical seven-fold comparator is the appropriate pre-GTZAN locked reference; a new external test set or nested selection is needed to establish superiority.

5.4. Limitations and decisive next tests

Three limitations bound the current claim. First, SMC activations are backbone-level OOF, but the single 4F decoder is not selected in a nested outer loop. Second, CASM searches 30–300 BPM while the default DBN searches 55–215 BPM. Recent SMC analysis reports many references below that lower limit [4]; a matched-support DBN is therefore required before attributing an SMC difference solely to CASM’s ambiguity mechanism. Third, the present aggregate tables do not isolate the sparse graph, strength-versus-width ambiguity gate, or the two safeguards. A direct PLPDP baseline, fixed-period DP, component ablations, and per-corpus fall-back and meter-path acceptance rates are the decisive experiments. Track-paired bootstrap or randomization tests should accompany them; three training seeds or eight folds alone do not justify significance claims.

6. CONCLUSION AND FUTURE WORK

CASM revisits structured beat decoding as an inductive bias conditioned on each input, not as a return to a fixed global grid. It joins activation-supported events with locally centred segment potentials and deliberately relaxes those potentials when periodic evidence is ambiguous. Once globally calibrated, the same deterministic module runs without upstream retraining, user annotation, or per-track tempo

Table 2. Contextual comparison with recent beat/downbeat trackers. Values are percentages reported under each cited system’s protocol; differing training data, checkpoints, and inference procedures make this literature context, not a controlled ranking. Our rows use GTZAN three-seed means and SMC eight-fold OOF means.

Method	GTZAN (Beat / Downbeat)						SMC (Beat)		
	F1	CMLt	AMLt	F1	CMLt	AMLt	F1	CMLt	AMLt
HN-MERT w. DBN [36]	89.7	81.4	94.3	77.4	71.6	87.2	61.7	—	—
HN-MusicFM w. DBN	89.2	80.9	93.7	79.8	73.2	89.5	60.6	—	—
BeatFM-MERT w. DBN [37]	<u>89.5</u>	<u>82.7</u>	<u>93.9</u>	76.7	70.2	85.4	61.3	—	—
BeatFM-MusicFM w. DBN	89.1	80.6	93.5	<u>79.6</u>	<u>74.4</u>	<u>88.7</u>	60.5	—	—
BeatMamba w. DBN [38]	88.9	82.3	94.3	—	—	—	58.7	46.8	62.4
BeatKAN w. DBN [39]	88.2	78.1	92.3	—	—	—	59.8	48.4	<u>64.0</u>
MDM-BeatThis (direct) [29]	89.7	82.9	92.5	79.5	76.4	88.5	61.0	48.9	64.3
BeatThis (direct) [3]	89.1	79.8	89.8	78.3	67.2	79.1	<u>62.7</u>	<u>51.4</u>	61.0
BeatThis w. DBN	88.1	80.5	91.1	77.4	73.4	87.8	57.5	47.5	62.5
BeatThis w. CASM (proposed)	89.3	80.2	90.4	78.9	70.4	85.1	63.0	53.8	64.3

GTZAN rows from this work average three independently trained backbone seeds; SMC rows from this work average eight fold-level macro scores from 217 track-disjoint OOF predictions. These are different descriptive replication axes. CASM uses the same Frozen-4F decoder in both cases; its SMC row is backbone-level OOF but not strict nested OOF for decoder selection.

and meter settings. Current evidence supports a conservative F1–continuity alternative to Direct and DBN decoding, not universal dominance. The next version should use strict nested or leave-one-dataset-out calibration, matched tempo-support baselines, PLPDP and mechanism ablations, and explicit safeguard statistics. Multi-hypothesis period paths, causal decoding, and a differentiable duration layer are natural extensions, as is evaluation on every recent tracker that releases compatible activations.

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